



Space Systems Implementation

Gas-Surface Interactions in Passive Intakes for Air Breathing Electric Propulsion

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1 Introduction

The development of Very Low Earth Orbit (VLEO) has long been a subject of interest for the electric space propulsion community. Generally, VLEO refers to mean orbits between 100 km and 450 km [1]. Within this broad range of orbits, the atmospheric density and with it the drag forces acting on the spacecraft substantially increase with lower orbital altitudes. These drag forces must be counteracted to maintain orbit, which is the reason why mission planners usually avoid operating in these denser regions. However, for electric propulsion (EP) devices, this environment creates unique opportunities. While the low thrust of EP systems limits rapid maneuvering capabilities, it perfectly fits the requirements of VLEO station-keeping. In fact, an electric thruster can be fired continuously throughout the mission time span to effectively compensate the counteracting drag forces. Through this, a well-designed system can enable missions that run for months or years, as successfully demonstrated by ESA's GOCE mission [2].

In recent years, interest in VLEO is particularly driven by the concept of Air-Breathing Electric Propulsion (ABEP). While GOCE was by all means successful, not many missions justify the additional drag and fuel costs for flying 250 km closer to Earth. Now, the ABEP concept re-envision missions in VLEO by removing the additional fuel from the equation and instead aiming to use the residual atmosphere as fuel. A fully developed VLEO platform could eliminate many of the disadvantages of flying in the thermosphere and enable new scientific and economic opportunities. For this to work, two main tasks need to be fulfilled: For one, a structure has to be developed that collects and compresses the residual gas and makes it available to the propulsion system. For another, the electric propulsion system needs to be optimized for the new type of fuel [1].

Fundamentally, a complete knowledge of the respective VLEO environment is needed to understand the input parameters for the system. To provide sufficient propellant availability for the EP system, possible orbits need to lie below 250 km. Notably, atmospheric density in this region is driven by the solar radiation causing significant variations between day and night sides of the orbit. Due to the high orbital velocity, the atmospheric gases (predominantly atomic oxygen and nitrogen) impact the satellite at more than 7 km s^{-1} . Thus, the aerodynamic system can be described by hyperthermal free molecular flow, a regime where the bulk velocity of the gas significantly exceeds its own thermal velocity. Furthermore, the mean free path length is large compared to the extent of the satellite's aerodynamic surfaces making the flow effectively collision-less. Consequently, this regime can only be modelled by simulating individual gas particles. As a result, an accurate description of the gas-surface interaction (GSI) between the atmosphere and the satellite is essential to model the system [3].

The design and optimization of the intake structure is a critical area of research in the field of ABEP. Recent studies have simulated and optimized the geometry of the intake utilizing the Direct Simulation Monte Carlo (DSMC) method [4–6]. While these studies successfully implemented widely available open-source DSMC codes, vastly different optimal geometries were found depending on the assumed GSI kernel. The structures ranged from simple parabolic geometries to honeycomb

intakes with linearly narrowing profiles. This highlights the need for experimental data for the given atmospheric environment to constrain the parameters for the underlying GSI model.

At the Ion Thrusters working group at Justus-Liebig-University Gießen, two experiments are solely focussed on gathering experimental data on the scattering behavior in VLEO-like environments. One experiment aims to directly measure the scattering kernel by using a highly collimated ion beam. This beam is directed at a sample surface and the reflected ions are measured with a Faraday cup scanning a 180° arc around the point of incidence. The second experiment tests a scaled-down parabolic intake in a wide ion beam emitted by a Radiofrequency Ion Thruster (RIT). By precisely determining the beam parameters and measuring the compression ratio achieved by the intake, this experiment can be compared to simulations with different GSI parameters. Identifying the simulation that best fits the experiment indicates the correct range of parameters. This report reviews these two experiments in detail and presents preliminary results.

The report is structured as follows: First, gas-surface interactions are reviewed in detail. This includes theoretical models of scattering kernels and sputtering. Sputtering is addressed due to the higher particle energies present in the experimental setup compared to the energies expected in VLEO. The following two chapters are dedicated to the experiments mentioned above. They each explain the experimental setup, describe the measurement procedures, discuss the results, and finally outline future steps. Lastly, the findings of the work presented in this report are summarized and placed in the academic context.

2 Modelling Gas-Surface Interactions

In continuum fluid dynamics, gas-surface interactions (GSIs) are handled on a macroscopic level. Here, surfaces are treated just like the boundaries that confine the simulation domain. At these boundaries, the state variables must comply with a set of constraints that aim to represent the physical reality. Usually, the fluids normal velocity at the boundary is set to zero, so that particles cannot penetrate the surface. Moreover, the tangential velocity can also be constraint or set to zero (no slip boundary). The formation of boundary layers, shock layers, and flow dynamics then follows directly.

The hyperthermal free molecular flow that governs the VLEO environment requires the simulation of individual gas particles. At a solid surface, these particles encounter a barrier at which they are scattered. Physically, the GSI comes down to the microscopic interaction of the atomic/molecular electron shell with the surfaces electronic structure. Some approaches, like the Soft Cube or Hard Cube Model, describe this by simplified elastic collisions of rigid particles with appropriate atomic surface potentials. Another, widely applied method, models GSIs probabilistically by defining scattering kernels that produce a scattering cloud for a given incoming particle trajectory. The latter is reviewed in detail in section 2.1 [3].

Lastly, particles that have sufficient kinetic energy (several dozen electron volts) can overcome the surface binding energy, enter the crystal lattice of the surface material, and sputter target atoms out of it. At orbital velocities, sputtering is not present because particles have about 5 eV kinetic energy. In the experiments presented in this report, impact energies reach several hundred eV, making sputtering a prevalent phenomenon. To take care of this discrepancy, section 2.2 presents simulation results to approximate the impact of sputtering on the following experiments.

2.1 Scattering Kernels

It is helpful to examine in detail what is happening during the scattering of a single gas particle. The incoming atom/molecule has an incidence velocity vector ξ_i that can be decomposed into a normal and a tangential component to the target surface. At one point, the particle will strike the surface at position $\mathbf{x}_i$ at time t_i . It is important, that the surface atoms are not locked in place but rather oscillate depending on their thermal energy. Furthermore, the surface is generally not perfectly flat but rather rough and bumpy. Thus, particles do generally not bounce off the surface like billiard balls but also experience energy and momentum transfers. These energy and momentum transfers depend on gas and surface properties and significantly shape the particle's outgoing velocity vector ξ_r [3].

Scattering kernels provide a probabilistic correlation between the incidence and reflected velocity. Generally, it is assumed that the residence time on the surface is small (no absorption) and that the particles do not change position between impact and reflection (no diffusion and no changes to the surface structure). Thus, the scattering kernel takes the general form

$$P(\mathbf{x}_i \rightarrow \mathbf{x}_r, t_i \rightarrow r_r, \xi_i \rightarrow \xi_r) = K(\xi_i \rightarrow \xi_r). \quad (1)$$

Since the kernel describes a probability density, it must not be negative and must fulfill the normalization condition

$$\int_{\xi_{r,n}>0} K(\xi_i \rightarrow \xi_r) d\xi_r. \quad (2)$$

Moreover, the process needs to be reversible, so that

$$f_0(\xi_i)|\xi_{i,n}|K(\xi_i \rightarrow \xi_r) = f_0(\xi_r)|\xi_{r,n}|K(-\xi_r \rightarrow -\xi_i), \quad (3)$$

where $f_0(\xi_i)$ and $f_0(\xi_r)$ are the velocity distributions of the incidence and reflected particle respectively [7].

An important parameter for GSI are the so-called *accommodation coefficients*. They take values between 0 and 1 and describe the degree to which a flow of gas exchanges kinetic or thermal properties during interaction with a surface. These parameters depend on the properties of the gas and the surface and must be determined experimentally. The general form of an accommodation coefficient is

$$a = \frac{\Phi_i - \Phi_r}{\Phi_i - \Phi_w}, \quad (4)$$

where Φ_i , Φ_r are the incidence and reflected fluxes of the given property, and Φ_w is the flux that corresponds to full accommodation. From this, coefficients can be defined for all relevant fluxes for a given interaction model [8].

In 1879, Maxwell [9] came up with a scattering description that is still widely used today. He built the now called *Maxwell Model* from two extreme cases of gas-surface interaction. According to this theory, σ particles undergo a specular reflection, where $\xi_{r,n} = -\xi_{i,n}$ and $\xi_{r,t} = \xi_{i,t}$ for both tangential components. The other $(1 - \sigma)$ particles are assumed to stay in sufficiently long contact with the wall, so that they reach thermal equilibrium. Those particles are then diffusely reflected from the wall according to a Maxwell-Boltzmann velocity distribution (see Figure 1). Maxwell then defined the complete scattering kernel by a linear combination of the two extreme cases outlined above:

$$K_M(\xi_i \rightarrow \xi_r) = (1 - \sigma)\delta(\xi_i - \xi_{r,\text{specular}}) + \sigma f_M(\xi_r)|\xi_{r,n}|, \quad (5)$$

where $\xi_{r,\text{specular}} = \xi_r - 2(\xi_r \cdot \mathbf{n})\mathbf{n}$, i.e. the same as the incidence vector with a flipped normal component ($\mathbf{n}$ is the surface normal). Moreover, σ is the accommodation coefficient that determines the probability of a diffuse reflection and does not represent a ratio of fluxes as defined above [8]. This simple model covers the main GSI effects and was thus widely applied to DSMC simulations. However, experiments showed that this diffuse scattering with a single specular peak is not physical and that the true probability distribution is rather petal shaped [3].

Almost one hundred years later, in 1971, Cercignani and Lampis [7] revisited the problem of gas-surface interactions from a more theoretical perspective. They developed the underlying kernel theory presented in the second paragraph of this section and used it to derive a scattering law that agrees very well with modern experiments (*CL Model*). Later, in 1991 [10] and 1995 [11] Lord expanded the theory to cover diffuse scattering with partial energy accommodation, partially diffuse scattering,

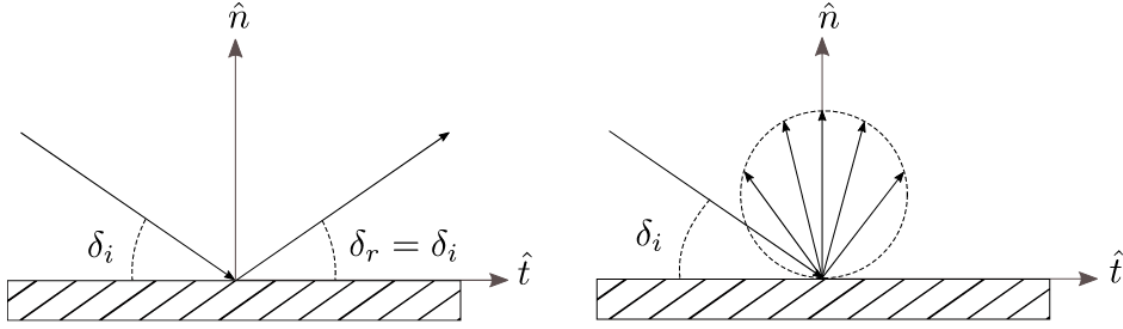


Figure 1: In this schematic from [3], the two extreme cases of scattering are depicted. On the left, specular reflection is shown, for which only the normal component of the incidence velocity flips sign, while the others remain the same. On the right, a completely diffuse reflection is visualized that requires the incidence particle to get into full thermal equilibrium with the surface before reflection.

and the vibrational energy of diatomic molecules as an anharmonic oscillator. Furthermore, he adapted it for use in DSMC simulations. Thus, this model is today mostly referred to as the *Cercigiani-Lampis-Lord (CLL) Model*. For an isotropic surface the scattering kernel for the normal component ξ_n and the two tangential components ξ_t can be written as:

$$K_{\text{CLL}}(\xi_{i,n} \rightarrow \xi_{r,n}) = \frac{2\xi_{r,n}}{\alpha_n} I_0 \left[\frac{2(1 - \alpha_n)^{1/2} \xi_{r,n} \xi_{i,n}}{\alpha_n} \right] \exp \left[-\frac{\xi_{r,n}^2 + (1 - \alpha_n) \xi_{i,n}^2}{\alpha_n} \right], \quad (6)$$

$$K_{\text{CLL}}(\xi_{i,t} \rightarrow \xi_{r,t}) = \frac{1}{\sqrt{\pi \sigma_t (2 - \sigma_t)}} \exp \left[-\frac{[\xi_{r,t} - (1 - \sigma_t) \xi_{i,t}]^2}{\sigma_t (2 - \sigma_t)} \right], \quad (7)$$

where I_0 is the modified Bessel function of the first kind and zeroth order, α_n is the normal energy accommodation coefficient, and σ_t is the tangential momentum accommodation coefficient [10]. This model produces exactly the petal shaped scattering distributions also seen in experiments. However, the two accommodation parameters are heavily dependent on the surface material, surface texture, gas type, and flow velocities. Thus, they cannot be predicted by theory and must instead be determined by dedicated experiments.

2.2 Sputtering

Sputtering describes the erosion of solid surfaces under ion bombardment. Ions of sufficient energy penetrate into the outer surface layers of the solid and transfer their energy to the target atoms by collisions. These atoms recoil and generate other recoils, triggering a collision cascade. Some recoils are directed backwards and reach the surface with enough energy to escape the solid. The erosion rate is quantified with the *sputtering yield* which represents the number of emitted atoms for each incident ion. This yield depends on the target properties and ion beam parameters [12].

Estimates for the sputtering yield expected for the experiments below are generated using the SRIM software package from Ziegler and Biersack [13]. Besides

other functionalities, SRIM has a tool that simulates the Transport of Ions in Matter (TRIM). In this tool, one can define an incoming ion beam (ion species, beam energy, angle of incidence), as well as the target material (layers, atomic composition, density). The software then generates an amorphous representation (no lattice structure) of the defined target material and performs Monte Carlo simulations for the interactions of the incidence ions with this material. This way, the simulation tracks the trajectory of every single ion and the resulting cascade of its displaced target atoms. Any target atom reaching the surface with an energy exceeding the surface binding energy is classified as a sputtered atom. Using the results from N simulated particles, the sputtering yield and spatial probability distribution (analogous to the scattering kernel) can be calculated from the accumulated statistics.

In the following, TRIM simulations are evaluated for the surface materials and ion beams considered in the experiments below. The full intake structure is made from stainless steel and incoming ions impact the inner surface at roughly 70° to 80° in respect to the surface normal. Since ions will generally collide multiple times with the structure, other impact angles will also occur. Consequently, sputtering yields are calculated for all possible incidence angles $\alpha = [0^\circ, 90^\circ[$ in respect to the surface normal. Furthermore, the beam energy is varied to simulate different particle velocities. In general, the beam energies are much higher than the 5 eV expected in VLEO because the extraction grids of the utilized ion sources (Radiofrequency Ion Thrusters - RIT) are optimized for beam energies of more than 1000 eV. Still, proper adjustment of the extraction grid voltages can produce beam energies as small as 200 eV. Therefore, the sputter simulations are performed for three exemplary energies $E = \{200 \text{ eV}, 400 \text{ eV}, 1000 \text{ eV}\}$. For now, the experiments are performed with argon and krypton since those noble gases are easier to handle.

The results provide an important understanding of the secondary sputtering effect that will occur during the scattering experiments. In Figure 2, the sputtering yield for both gases at energies E is plotted against different incidence angles. Clearly, lower energy ions cause significantly less sputtering. Also, the sputtering yield shows a different angle-dependency for argon (40 u) and krypton (84 u). While krypton has a pronounced peak at around 70° for all energies, the peak for argon lies lower and is less prominent. In fact, at 200 eV the sputtering yield for argon is almost constant for all angles. The angular sputtering probability distributions look very similar to the Maxwell scattering kernel in section 2.1, although it describes a different physical phenomenon. For smaller energies, the ions only penetrate into the upper layers of the material so that the recoil atoms do not need many collisions to reach the surface. Thus, they keep a lot of the incidence ion's tangential velocity, producing a probability distribution with a clear peak around 60° . However, for higher energies, the ions have enough energy to enter deeper into the material so that the sputtered atoms are the result of a longer collision cascade. Consequently, the directionality of the initial ion is lost, and the target atoms leave the material in almost random directions.

This analysis reveals that accounting for sputtering effects is essential in the subsequent experiments. For the controlled scatter experiment, it is important that after the scattering process a measurable ion current remains detectable across the angular distribution. Fig. 4 presents the scattering probabilities for argon and kryp-

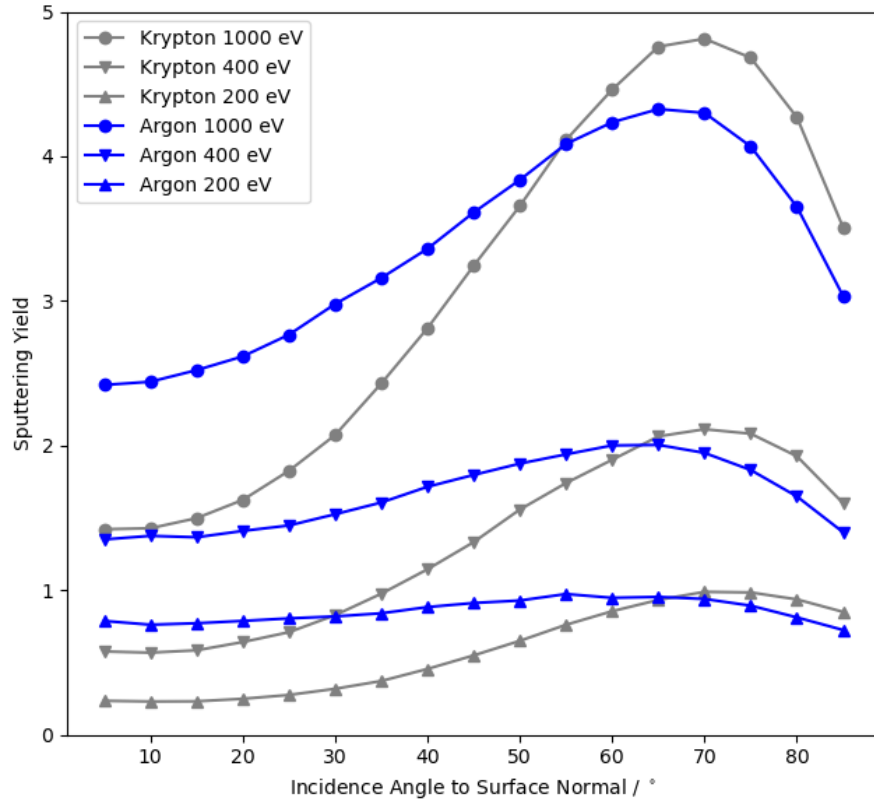


Figure 2: This plot compares the sputter yield of stainless steel for argon and krypton at different impact energies. Higher energies lead to much higher sputtering. Moreover, argon's sputtering yield shows less dependence on the incidence angle.

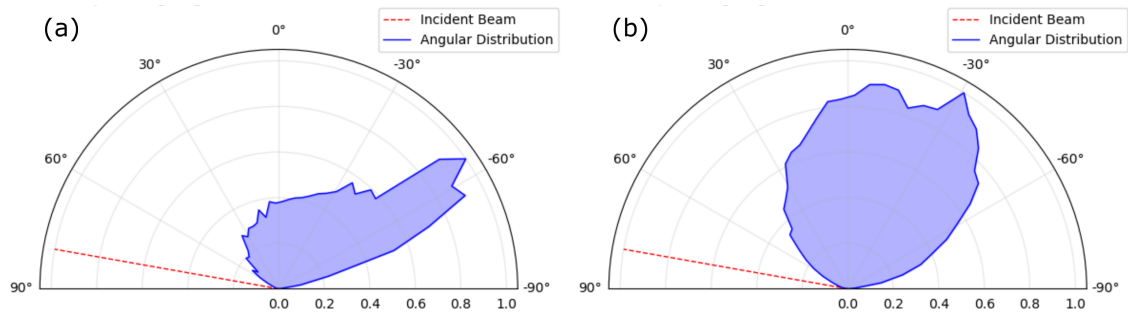


Figure 3: These plots compare the angular sputtering distribution at an incidence angle of 10° . (a) is the distribution resulting from a 200 eV beam. It has a random component of particles that are sputtered independent of the incidence angle but a large contribution of particles with a preferred directionality. In contrast, the sputtering distribution (b) from ions at 1000 eV shows almost no directionality.

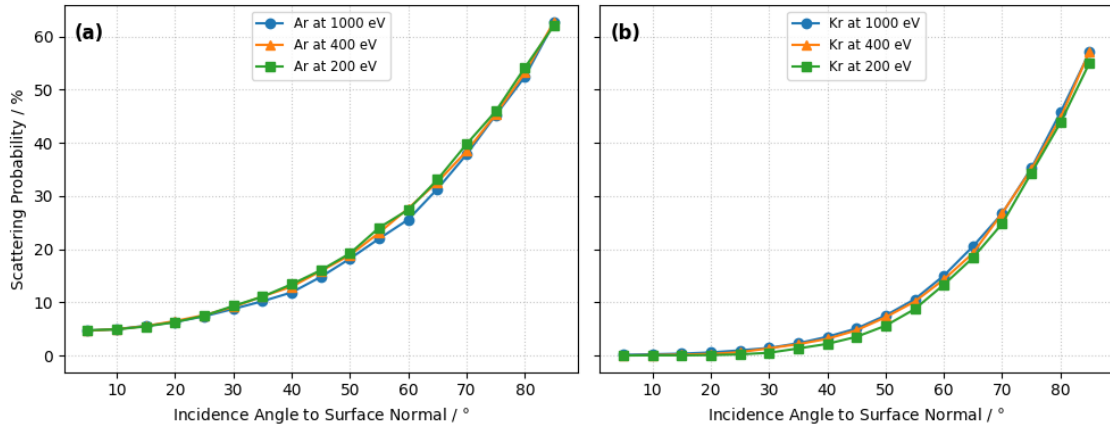


Figure 4: Here, the probability for an impacting particle to scatter at the surface and not enter the material structure is depicted. For small incidence angles to the surface normal, the particles impact the surface with a large normal velocity component and are thus likely to enter the target surface. Comparing the scattering probabilities for (a) Argon and (b) Krypton reveal that Krypton's scattering probability is 4.5 % to 10 % smaller than Argon's.

ton. Particularly at small angles of incidence, argon is the superior gas choice with a scattering probability approximately an order of magnitude higher than that of krypton. In the full intake experiments, the sputtered atoms increase the number of particles collected in the tank and therefore distort the pressure measurement. To address this, a collaborating research group from South Korea accounts for sputtering in their DSMC simulations that will be compared to the experiment. Still, ion energies should be kept as small as possible to more closely replicate VLEO energies and minimize sputtering. Argon is also preferred due to its relatively constant sputtering yield across various angles of incidence, which simplifies the modeling of sputtering effects.

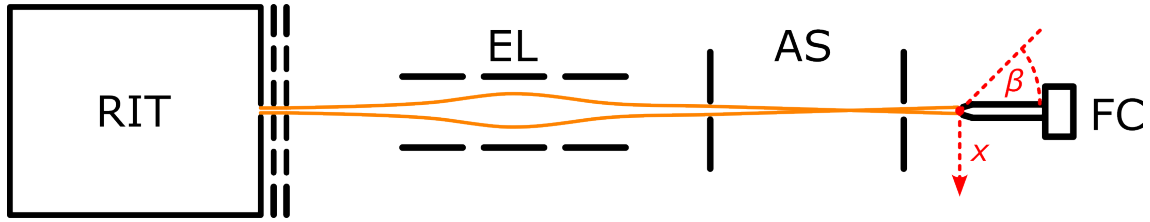


Figure 5: This schematic describes the experimental setup used during the beam characterization. From left to right it consists of an ion source (RIT), einzel lens (EL), an aperture system (AS) consisting of two apertures, and a Faraday cup (FC). The latter sits on a rotary stage that is mounted on a linear stage enabling both rotation (by β) and translation (by x) of the cup assembly.

3 Scatter Experiment

In this experiment, the scattering probability distribution for different materials is measured directly. To achieve this, an ion beam is collimated using ion optics ensuring that a parallel beam with a small spot size impinges on the target. At the target, the beam particles strike the surface at approximately the same location with a constant angle of incidence. Governed by the underlying probability distribution, the particles will scatter in different directions and are subsequently detected by a Faraday cup. It is imperative that a sufficient fraction of the incidence ions scatter towards the cup. Before measuring the scattering distribution, the incident beam needs to be characterized properly. The respective measurements are described and discussed in the following.

The described experiment is topic of Mathis Reuß-Hennschen’s master’s thesis. He prepared the experimental setup and acts as the project lead. The author of this report assisted with the assembly and alignment of the experimental setup, helped to install the assembly in the vacuum chamber, and assisted in establishing the electrical connections. Furthermore, the author performed full system checks after the chamber was evacuated, and contributed equally to the testing and measurement procedures for the beam characterization. However, the author was not present during the perveance measurements. At the time of writing, the full scatter experiment is pending due to a lack of available vacuum test facilities.

3.1 Experimental Setup

The heart of the experimental setup is the ion source that produces the ion beam. Building on the expertise of the working group, a RIT-10 is used for that purpose. This thruster utilizes an extraction grid system (usually three successive grids) with numerous holes arranged within a diameter of 10 cm. Through these holes, ions are accelerated out of an inductively coupled plasma, creating a wide exhaust plume. For this experiment, the first grid is replaced by an aperture with a single hole in the center so that the source emits a single narrow beam.

Right behind the thruster, an electric einzel lens is used to focus the narrow beam. This device is commonly used in ion optics. It consists of three successive metal cylinders that are electrically isolated from each other. The two outer cylinders are

usually grounded, while the center one is put on a positive potential. The resulting electric field structure has a focusing effect on the ion beam.

Next, two apertures with a diameter of $D = 5$ mm further collimate the beam. They are placed $d = 14$ cm apart from each other. The first aperture has a distance of 7.9 cm to the einzel lens, while the second has a distance of $d_{\text{cup}} = 6$ cm to the rotary stage's axis of rotation. Only ion trajectories that pass both apertures reach the target surface that is placed behind the last aperture. This limits the maximum theoretical divergence to about

$$\alpha = \arctan\left(\frac{D}{d}\right) = 2.05^\circ. \quad (8)$$

Figure 5 shows a schematic depiction of the setup used for the beam characterization. As mentioned above, the target surface will be placed behind the last aperture to measure the scattering distribution. However, for the characterization of the incident beam, the target surface is exchanged for a rotary stage that is mounted on top of a linear stage. This allows the Faraday cup to be rotated and translated inside the vacuum chamber.

For precise measurements, a long collimator is added to the Faraday cup. It has an opening diameter of 3 mm and extends 10 cm from the opening of the Faraday cup itself. Therefore, the cup only measures ions with minor angular deviations from the normal direction. This allows for a high angular resolution. Moreover, the opening of the collimator is positioned such that it lies on the axis of rotation of the rotary stage. Thus, the translational and rotational position of the opening can be controlled separately.

Finally, all components are mounted on an extruded aluminum profile with a (40x80) mm cross-section. The thruster is mounted on a fixed stand on one end of the profile and is used as a reference for alignment of the remaining pieces. Custom aluminum brackets are designed for the einzel lens and the apertures to allow them to move freely in two dimensions. Alignment is performed using a reference laser that is directed towards the thruster from the opposite side of the profile. By using a mirror, the laser-thruster system is brought into alignment. The remaining components are aligned along the resulting axis.

3.2 Measurements

First, it is important to find operating points for the extraction grid system of the RIT-10. The extraction dynamics are determined by the voltages applied to the three grids (in order: screen, acceleration, and deceleration grid). Here, the screen grid sets the beam energy and is used to bias the whole plasma to a positive potential V_{PHV} . Next, the acceleration grid extracts the ions out of the plasma with a slightly negative potential V_{NHV} . Lastly, the deceleration grid is grounded to prevent ion backstreaming. All three grids collectively govern the ion exit trajectories. Using simulations, extraction grid systems for RITs are usually optimized for PHV voltages of 1 to 2 kV and NHV voltages of 100 to 300 V. However, by using different combinations of PHV and NHV voltages, the beam energy can be reduced significantly.

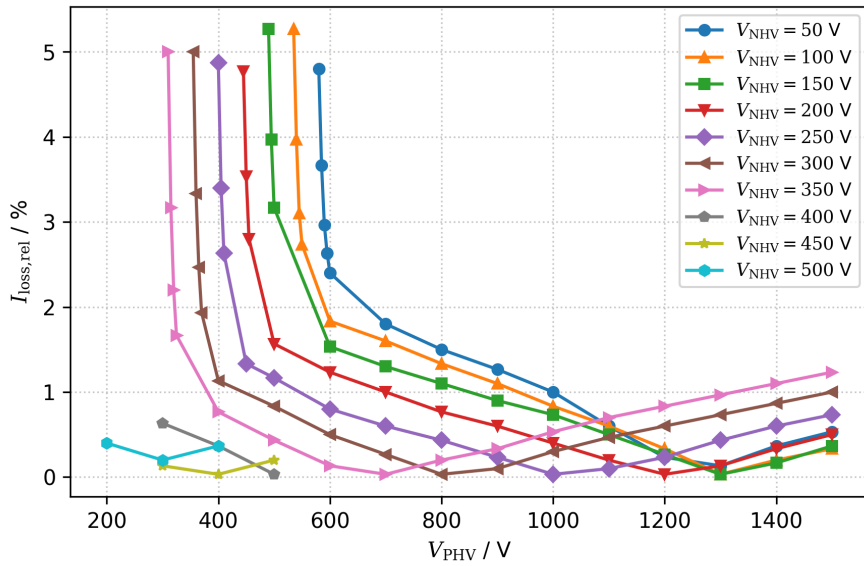


Figure 6: For this plot, sweeps of the PHV voltage are performed for different NHV voltages. All sweeps show a clear minimum - the optimal extraction energy for the given NHV voltage - and a sharp rising edge towards small PHV voltages, indicating the loss of focus and a considerable amount of ions hitting the acceleration grid. Higher NHV voltages clearly allow for smaller PHV voltages (i.e. lower beam energies). The curves for NHV voltages from 400 to 500 V were not captured over the whole range of PHV voltages but only around the minimum.

When lowering the PHV voltage, it is important to keep track of the relative loss current. The relative loss current is the ratio of the acceleration grid current to the total beam current and thus correlates with the percentage of ions hitting the acceleration grid. For well-designed and well-operated extraction systems, charge exchange (CEX) ions formed in between the grids are the main source of this loss current. However, the relative loss current can also be used to detect poor focusing of the beam, when the ion trajectories directly hit the acceleration grid. Consequently, a series of perveance measurements is performed that track the relative loss current for successively lowered NHV and PHV voltages. Plotting the relative loss current against V_{PHV} for different V_{NHV} reveals operating points that allow for low beam energies while keeping the relative loss current manageable (see Fig. 6).

Next, the beam current behind the apertures is measured without active focusing with the einzel lens. For this, the Faraday cup is moved slowly on the linear stage while recording the current. With this, a trapezoidal current profile is detected that linearly increases at the edge of the aperture, remains constant in the middle and falls off linearly on the other side of the aperture again (see Fig. 7a). Using this, the center axis of the experimental setup is defined as the center of the trapezoid profile. In the following, the Faraday cup is positioned in the center and the rotary stage is slowly turned. The rotary stage is calibrated to 0° at the position where it detects the highest current.

In the following, the einzel lens is turned on and the voltage is varied. The Faraday cup remains in its calibrated position and is used to measure the current

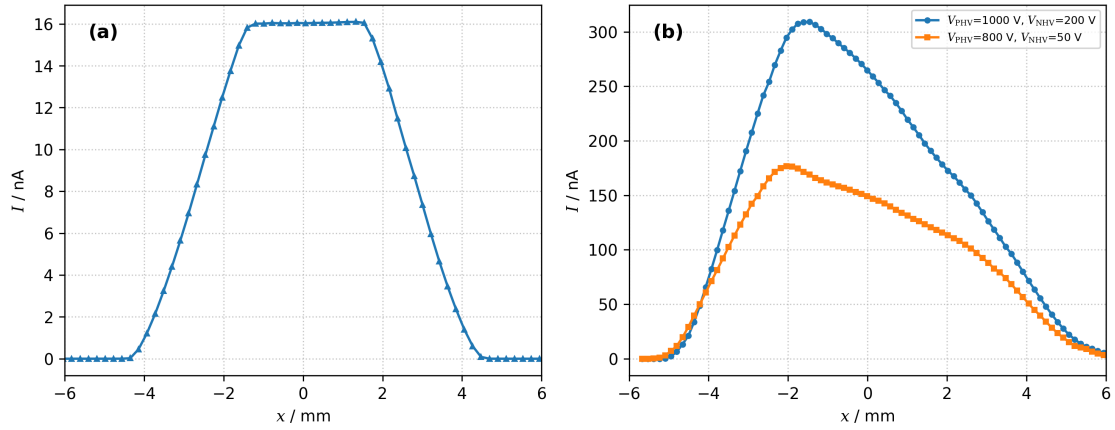


Figure 7: Linear beam profiles for different operating points. (a) shows the profile for a grounded einzel lens for a PHV voltage of 1000 V. The beam is unfocused so the profile corresponds to a cut-out of the divergent beam. (b) compares the profiles for a 1000 eV and a 800 eV beam with an activated focusing lens. Both profiles show a shifted maximum and asymmetric shape. The lower energy beam has a considerably lower current maximum (about half).

for the different focus voltages. First, a rough scan with large step sizes is performed. After the peak is roughly located, the scan is continued more precisely, with smaller step sizes. The final result reveals the ideal focus voltage of $V_{\text{lens}} = 770$ V for a beam energy of 1000 eV as shown in Fig. 8. For other beam energies, this process is repeated to find the respective ideal focus voltages.

Subsequently, a linear beam profile is recorded by moving the Faraday cup through the beam at 0° . Fig. 7b compares the profiles of the focussed beams for two different energies with the profile without focusing. First, the maximum current increases significantly from 16 nA to 175 and 310 nA respectively. Second, the beam shape changes drastically from a flat profile to an almost triangular shape with a shifted maximum. Both maxima lie approximately 2 mm left of the center.

Lastly, the linear and rotational measurements are combined to reveal the proper beam profile. For this, the cup is rotated from -9° to 9° for ten different positions of the linear stage. Fig. 19 contains two plots that portray the same data differently. (a) shows the one-dimensional angular profile for the different positions. From this, the symmetric shape for the profiles to the right of the maximum and their successive shift towards larger angles becomes visible. The two profiles left of the maximum are asymmetric and have a broader angle distribution. (b) depicts a two-dimensional color map plot from which all described trends are visible, except for the shape of the angular profiles.

3.3 Discussion

The perveance measurements show the expected physical behavior. In fact, the sharp edges towards smaller PHV voltages indicate that the ion trajectories directly impact the acceleration grid. Furthermore, higher NHV voltages allow for smaller PHV voltages (lower beam energies). However, the voltage settings that are needed

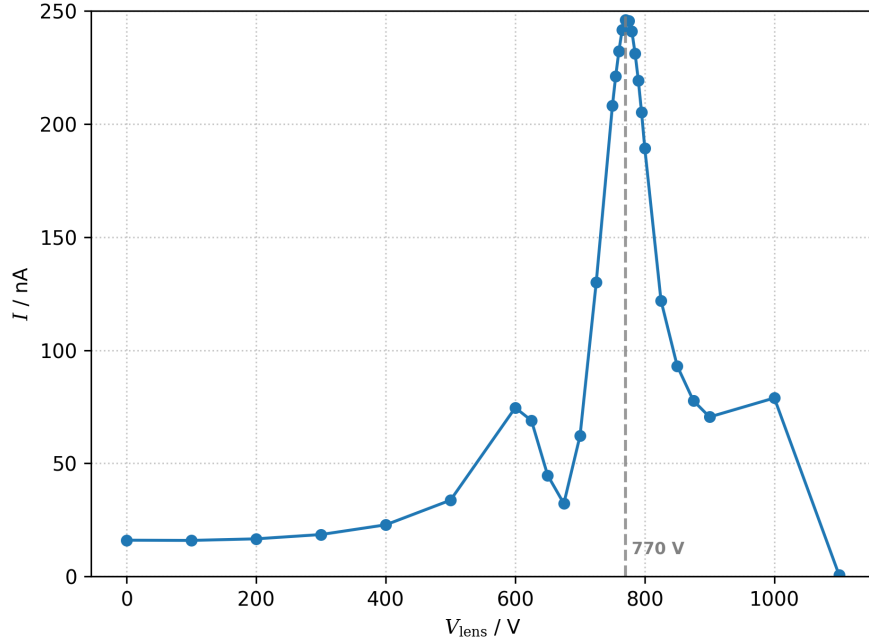


Figure 8: Exemplary voltage sweep to find the optimal einzel lens voltage. Around 770 V a clear maximum emerges.

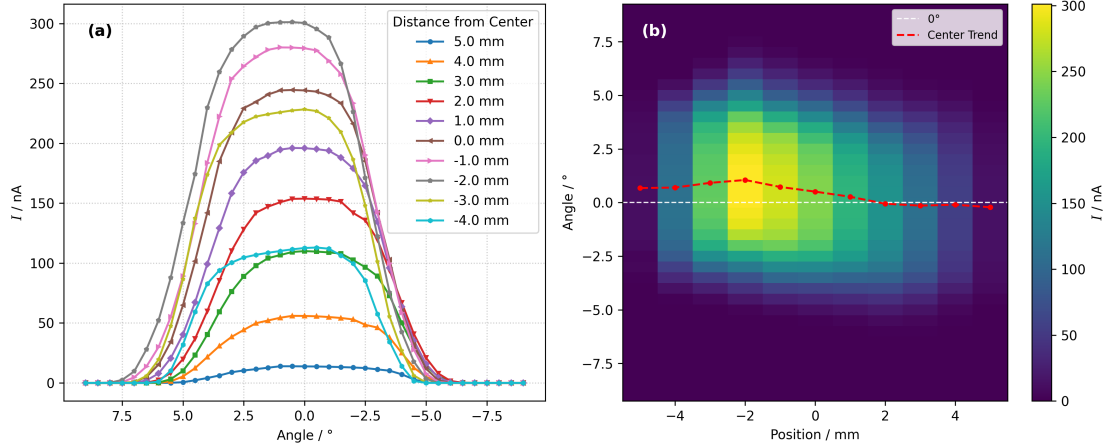


Figure 9: Two-dimensional beam profile measurement for $V_{\text{PHV}} = 1000 \text{ V}$ and $V_{\text{NHV}} = 200 \text{ V}$. (a) Angular beam profiles for different positions in the beam. The profiles to the right of the maximum are symmetric, the ones to the left asymmetric. (b) Combined plot depicting trends in angle and position. It shows that the current points increasingly to the left towards the maximum. In accordance with the linear profile the maximum lies about 2 mm left of the center.

Table 1: Optimal operating points for the different beam energies determined with the perveance measurement. $I_{\max}$ refers to the maximum current measured behind the aperture system.

$V_{\text{PHV}} / \text{V}$	$V_{\text{NHV}} / \text{V}$	$I_{\text{loss,rel}} / \%$	$I_{\max} / \text{nA}$
1000	250	0.03	331
900	300	0.1	258
800	300	0.03	191
700	350	0.03	141
600	350	0.13	105
500	400	0.2	85
400	450	0.03	60
300	450	0.13	35
200	550	0.2	19

to reach low beam energies have the consequence of producing a more divergent ion beam. That is the reason why the current profile for the 800 eV beam lies significantly lower than that for the 1000 eV beam. All suitable operating points are listed in Tab. 1 with their respective loss currents and maximum current behind the aperture system. Beam divergence has a significant effect on the maximum usable current. The following measurements were performed before this perveance measurement for two different operating points: $V_{\text{PHV}} = 1000 \text{ V} / V_{\text{NHV}} = 200 \text{ V}$ and $V_{\text{PHV}} = 800 \text{ V} / V_{\text{NHV}} = 50 \text{ V}$. The results provide an in-depth understanding of the beam characterization measurements and will be used to optimize the characterization of the listed operating points.

Next, the linear beam profile for the inactive einzel lens provides information about the geometry of the experimental setup. If the ion beam was perfectly parallel before entering the aperture system, the beam cut-out would be rectangular with a width of 5 mm. For a purely divergent beam, one would expect a flat top profile with a top width larger than 5 mm and steep slopes on both sides. Here, the first aperture would be irrelevant for the profile, since all trajectories that pass the second aperture would also pass the first one. However, the measured trapezoid profile has a flat top width of approximately 3 mm and shallow slopes creating a base width of about 9 mm. Since the ion source itself is divergent, an explanation needs to be found to understand the opposite effects of top width shortening and base width widening. First, one can observe that the maximum extent of the beam correlates with the maximum theoretical divergence calculated above:

$$D_{\max} = D + 2 \cdot \tan(\alpha) \cdot d_{\text{cup}} = 5 \text{ mm} + 2 \cdot \tan(2.05^\circ) \cdot 60 \text{ mm} = 9.3 \text{ mm} \quad (9)$$

This maximum divergence angle can only be reached when there is an ion trajectory passing through the first aperture at $x = 2.5 \text{ mm}$ with an angle of 2.05° , so it is barely able to pass the second aperture at $x = -2.5 \text{ mm}$, impacting the cup at $x = -4.6 \text{ mm}$. Yet, the divergent ion source cannot directly provide this inward pointing trajectory. One explanation would be that there are ions that are scattered off the inner walls of the inactive einzel lens, creating the missing inward trajectories.

Assuming this is true, one can also use this scattering theory to explain the shorter flat top width. In the center of the first aperture, there are straight, divergent, and convergent trajectories. Towards the edge of that aperture, however, there are only divergent and convergent trajectories. The divergent ones are clipped by the second aperture, so only the convergent (reflected) trajectories pass through to the Faraday cup, narrowing the perceived flat top profile. The most probable alternative explanation would be a combination of space charge effects, that could explain the base broadening and aperture misalignment, that could explain the shortened flat top width. Regardless, the axis symmetry of the current profile strongly suggests that the apertures are well aligned with each other.

After activating the einzel lens with the optimized focus voltage, a clear misalignment between the lens axis and the thruster and/or the aperture system becomes visible. Ideally, the maximum current would lie right in the center of the profile at $x = 0$. Instead, the maximum is shifted so far to the left, that only the profile to the right of the maximum is visible. The part on the left (and possibly the true maximum itself) is cut away by the second aperture so that an asymmetric shape emerges. Compared to the current profile without the lens, the base width increases by about 1 to 2 mm. This can be explained by space charge repulsion because the total charge flux through the aperture system is increased by at least one order of magnitude. To fix the asymmetry of the current profile, the alignment of the different components should be checked and fine-tuned. However, the different components would need to be aligned with a precision of less than one degree to achieve sub-millimeter precision for the focus point. Hence, the current alignment is sufficient and can later be corrected for during post-processing. After all, maximum current densities of 43.7 nA mm^{-2} and 25 nA mm^{-2} are obtained for the 1000 eV and 800 eV beam respectively.

Ultimately, the analysis of the angular measurements show trends that are in line with the linear profile but also show that the collimator is not very angle-specific. Ideally, the beam divergence results in ion trajectories pointing outward on both sides of the beam axis $x = 0$. However, given the observed shift in the linear profile, the out-of-axis focus causes a systematic angular bias towards positive angles (pointing leftwards). Indeed, the two-dimensional color map plot shows a shift to positive angles for the current-weighted center line of the angular profiles. Only very far to the edge of the aperture (and beyond) does the average angle point to the right. On the opposite side, left of the global maximum, the average angle unexpectedly gets smaller. This behavior is currently not understood but may be due to the general imprecision that is observed for the angular measurement. In fact, the large spread of the angular distribution that is present for all positions is unphysical. It is geometrically impossible for ion trajectories with large angles of 2.5° or more to reach the measurement point 4 mm right of the center. Thus, the collimator also transmits off-axis ions. These ions scatter off the inner walls and still reach the inside of the Faraday cup. This effect is negligible for the full scattering experiment, where different angles corresponds to separate spatial positions. However, for the current beam characterization, only the current-weighted average angle has physical significance.

4 Full Intake Experiment

In this experiment, a parabolic intake structure is placed in the path of a neutralized ion beam to simulate the flow conditions in VLEO. Two different fuel tanks will be placed at the exit of the intake, each equipped with a pressure gauge. While one tank structure is fully enclosed, the other has an opening of a known area in the back. A second pressure gauge will be mounted in front of the intake. Using the two pressure readings, the compression ratio of the intake can be calculated. Additionally, for the open tank structure, the mass flux through the intake can be derived. The results obtained through these measurements will be compared to DSMC simulations performed with different scattering kernels and various scattering parameters to find the underlying distribution that best describes the gas-surface interaction. In order to provide the simulations with accurate input data, various beam diagnostics are employed to determine the beam parameters. These beam characterization measurements are the topic of this section since the measurements are still ongoing at the time of writing.

The experiment described in this section is the topic of Max Reinhardt's master's thesis and the results will be part of Jana Zorn's dissertation. The author of this report supported the setup of the diagnostics in the vacuum chamber and assisted during the perveance, beam profile, and RPA measurements. Moreover, a Python module for analyzing the mentioned diagnostics was written and used to evaluate the measurements.

4.1 Experimental Setup

The experiment is set up in the JUMBO vacuum chamber, which has a total volume of 30 m^3 with a diameter of 2.5 m and a length of 6 m. With a pump performance of $150 \times 10^3\text{ L s}^{-1}$, a pressure of about $5 \times 10^{-6}\text{ mbar}$ is sustained during thruster operation. A water-cooled graphite beam dump reduces the wall-beam interactions. Moreover, the chamber features two separate 2-axis systems. The first one is positioned in the near-field of the thruster and can be moved in the XZ plane. The second one is positioned further away from the thruster to perform far-field measurements and can be moved in the XY-plane. A linear Faraday cup array in the z direction is mounted on top of it so that it can measure the beam current in all dimensions. The detailed setup is sketched in Fig. 10 and will be explained in the following paragraphs.

A RIT provides the ion beam that is neutralized with a radiofrequency (RF) neutralizer. The ion thruster has an extraction grid diameter of 15 cm of which the center of the third grid is used as the origin for the local coordinate system. On the same metal plate, next to the thruster, the neutralizer is positioned at an angle directed towards the exhaust plume. It emits electrons and is necessary to prevent charge build-up on satellites. For ground testing, a thruster could also be operated without a neutralizer, since any remaining charge could be conducted directly to ground. However, for this experiment the neutralized beam is desired to recreate the VLEO environment that consists of neutral particles. Although the ions and electrons do not recombine before they hit the intake, the net charge transferred to

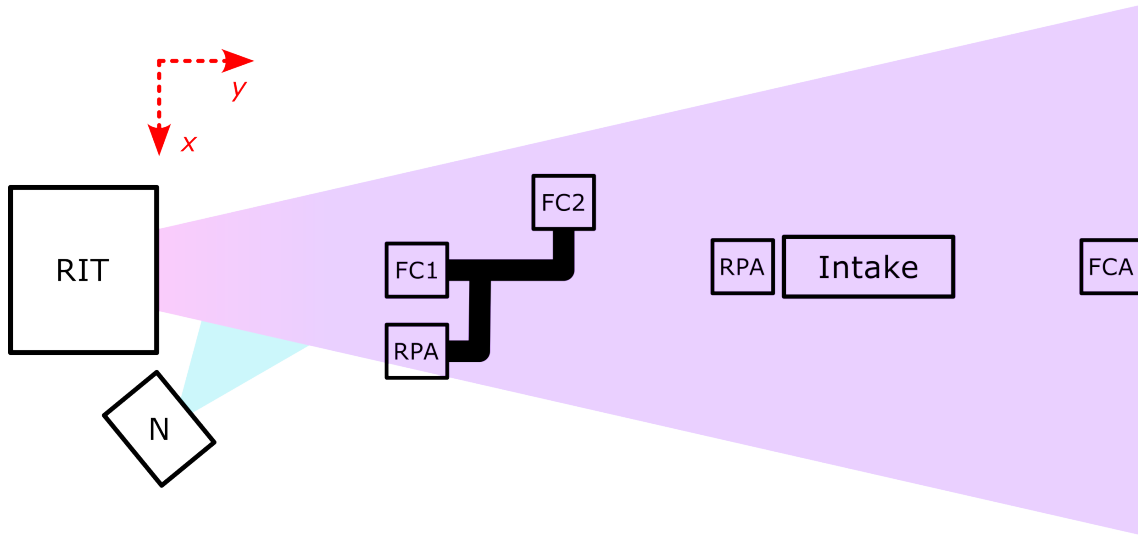


Figure 10: This schematic outlines the experimental setup used for the beam characterization of the full intake experiment. The ion source (RIT) and neutralizer (N) are mounted on brackets that stand on top of a metal plate. At a distance of 78 cm the scaled-down intake is positioned in the center of the beam. Directly below the opening of the intake, a retarding potential analyzer (RPA) is mounted with its opening facing upwards. Between the source and the intake a two-dimensional stage allows for movements in the x and z direction. Two Faraday Cups (FC) and another RPA are mounted on top of that stage. These probes face towards the thruster. Behind the intake, a Faraday cup array (FCA) is mounted on top of a stage that is movable in the x direction.

the intake is approximately zero.

At a distance of 303 mm from the thruster, a Faraday cup (FC1) is mounted on top of the XZ stage, which also contains two other diagnostic instruments. Another Faraday cup (FC2) is positioned 106.5 mm to the side and 218 mm behind the first cup. On the opposite side, a retarding potential analyzer (RPA) is placed 113 mm to the side of FC1 so that its opening lies 6 mm in front of the entrance hole of the cup. During beam characterization, the stage is kept at a constant height, so that the openings of all diagnostics lie at $z = 0$.

Behind the first 2-axis stage, the intake is placed on top of an extruded aluminum profile so that it is electrically floating relative to ground. Its opening has a distance of 780 mm to the thruster and is placed directly in the center of the beam at $x = z = 0$. Below the intake, another RPA is placed facing upwards to measure charge exchange ions. For the beam characterization measurements, no collection tank is placed behind the intake.

Lastly, the XY stage is placed in its home position 455 mm behind the intake. At the time of writing, no measurements have been performed with the Faraday cup array (FCA).

4.2 Measurements

As a first step, perveance measurements are performed for the RIT-15 thruster and RF neutralizer combination. As already described in the chapter detailing the

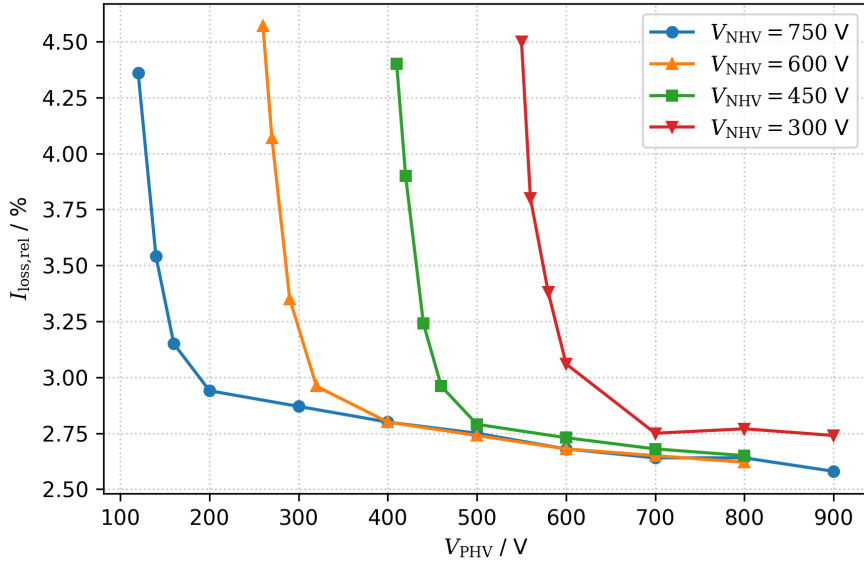


Figure 11: The relative loss current is measured for a series of different NHV and PHV voltages. Since only energies $\leq 900 \text{ V}$ are considered, only the part of the curves left of the maximum is visible. Lower NHV voltages consistently reduce the possible beam energies before the unfocused trajectories start to hit the grid.

scatter experiment, different NHV and PHV voltage settings are compared with respect to their respective relative loss currents. Higher NHV voltages generally allow for lower PHV voltages (i.e. lower beam energies) but also increase beam divergence. The acquired curves show the expected steep incline at voltages where the ion trajectories start to intersect with the acceleration grid. Suitable operating points are found between 200 and 1000 eV as depicted in Fig. 11.

In the following, the beam energy is measured externally using the RPA mounted on the first 2-axis stage. Although the PHV voltage already determines the approximate magnitude of the beam energy, the variable plasma potential adds uncertainty to the true energy of the beam. An RPA consists of a grid that is biased to a positive retarding potential and a Faraday cup that is used to measure the incident ion current. During a measurement, the retarding potential is varied, while the ion current is measured. When the retarding potential starts to repel incident ions, the ion current begins to fall off. This decline gets steeper until the most abundant ion energy is reached. Subsequently, the ion current reduces more slowly until the retarding potential is high enough to repel all incoming ions, and the current ideally drops to zero. The first derivative of the acquired current profile corresponds to the ion beam's energy distribution and generally has a Gaussian shape. The center of the Gaussian then corresponds to the mean energy of the beam, while the standard deviation describes the width of the energy distribution.

For each operating point, the RPA is placed in the center of the beam and a sweep of the retarding potential is performed. The resulting profiles meet the expected shape, although for some measurements an additional hump before the start of the steep decline is observed (see Fig. 12). This small increase in ion current is due to lensing effects induced by the RPA's grid system and is unproblematic since it

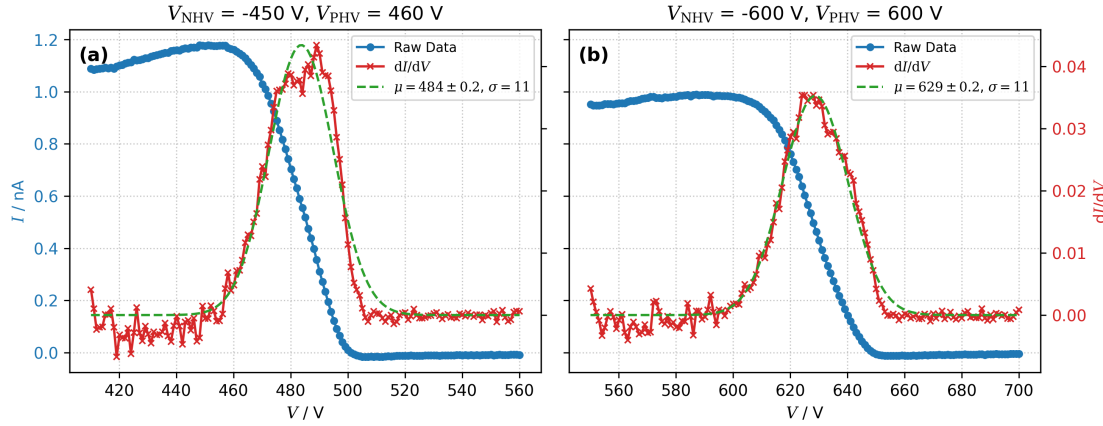


Figure 12: This plot compares two exemplary RPA measurement curves for different operating points. Both current profiles generally follow the Gaussian curve, although the curve in (a) exhibits a larger initial hump and features a flat part at the top of the peak. After correction of the measurement, the energies are determined to $462 \pm 4 \text{ eV}$ and $603 \pm 4 \text{ eV}$ respectively.

does not overlap with the relevant edge. For this reason, the first derivatives of the current profile mostly follow the expected Gaussian profile, so that the mean and standard deviations can be determined. In the final step, these results are corrected using a calibration curve. This curve was calculated by comparing the RPA's results to the known energy profile of an energy discrete ion source.

In addition to the beam energy, the beam divergence is an important quantity that changes the compression efficiency of the intake. In fact, it directly affects at what angles the ions impact the intake structure. For this measurement, the two Faraday cups that are mounted on the first 2-axis stage are used. The stage is positioned at $z = 0$ and moved through the beam from left to right. For each scan, the movement is subdivided into small steps so that the beam current is measured at 39 different positions. When plotting the measured current against its respective x-axis positions, two Gaussian profiles emerge for the two separate Faraday cups (see Fig 13a). By fitting a theoretical curve to the data, the standard deviations are obtained and used to calculate the beam divergence. However, before this final calculation, the experimental data needs to be corrected for divergence and charge exchange (CEX) effects.

First, the divergence correction is performed. This correction is necessary because the cups are moved linearly through the beam. In the center, where the beam divergence is zero, the ions fall straight through the aperture of the Faraday cup onto the collection surface. In contrast, towards the edge of the beam, the ions enter the aperture of the cup at an angle so that some ions do not reach the collection area. Therefore, the measured current is lower than the true current falling through the cup aperture. This can be corrected by approximating the divergence angle between the grid and the Faraday cup. In the near-field, it is common to use a *minimal angle* approach that sets the divergence directly in front of the thruster to zero [14, p. 56]. At the same time, for positions outside the thruster diameter, the tangential

distance to the thruster's edge is used to calculate the divergence correction angle

$$\beta = \arctan \left(\frac{\Delta x_{TEC}}{\Delta y_{TC}} \right), \quad (10)$$

where Δx_{TEC} is the tangential distance between the thruster edge and the cup, while Δy_{TC} is the normal distance between the thruster and the cup. Using

$$I_{\text{true}} = \frac{I_{\text{meas}}}{\cos(\beta)}, \quad (11)$$

the measurements are corrected. As a result, the further the distance between the measurement point and the thruster center, the larger the divergence correction.

Next, the CEX effects are taken into account. Charge exchange describes the process of charge transfer from a beam ion to a neutral background gas particle in the chamber. This charge transfer is only connected to a small transfer in momentum so that the neutralized atom continues on its trajectory, while the newly formed ion moves with its initial random thermal velocity. Thus, for every charge exchange event, one ion is lost from the Faraday cup measurement. This loss of current can be compensated by using

$$I_{\text{true}} = I_{\text{meas}} \frac{1}{1 - n \cdot \sigma(E) \cdot z}, \quad (12)$$

where n is the background gas particle density, $\sigma(E)$ is the energy-dependent cross section for CEX collisions and z is the distance the ions travel from the thruster to the cup [14, p. 58]. For this report, the experimental cross section measurements for Argon from Phelps [15] are interpolated by a logarithmic fit in the range $E \in [200, 1000]$ eV leading to the following relationship between cross section and beam energy:

$$\sigma(E) = (8.217 - 1.011 \cdot \log(E))^2. \quad (13)$$

The theoretical approach for $\sqrt{\sigma} \propto \log(E)$ is motivated by the semi-classical theory formulated by Rapp and Francis [16]. Furthermore, the ion energy used to calculate the CEX cross section is correlated to the PHV voltage using a linear fit to the RPA curve depicted above. Ultimately, all positions experience a correction towards higher currents, with the positions having the largest tangential distance to the thruster requiring the greatest correction.

The impact of the divergence and CEX correction on the beam profiles is presented in Fig. 13. The larger and less divergent profile corresponds to FC1. It peaks at around 223 mm, which was also determined to be the position for the linear stage at which FC1 is centered in front of the thruster. The second profile belongs to FC2 and appears shifted to the right by approximately 106.5 mm because of its tangential distance to FC1. This separation distance is also in line with the physical distance between the cups measured before evacuating the chamber. Moreover, the Gaussian shape for FC2 is significantly broader compared to FC1 due to the expected beam divergence. The differences between the profiles on the left and right are minimal but visible. In fact, the dotted horizontal supporting lines help to indicate the difference in height of the fitted Gaussian profiles. While the lifted tails expected by

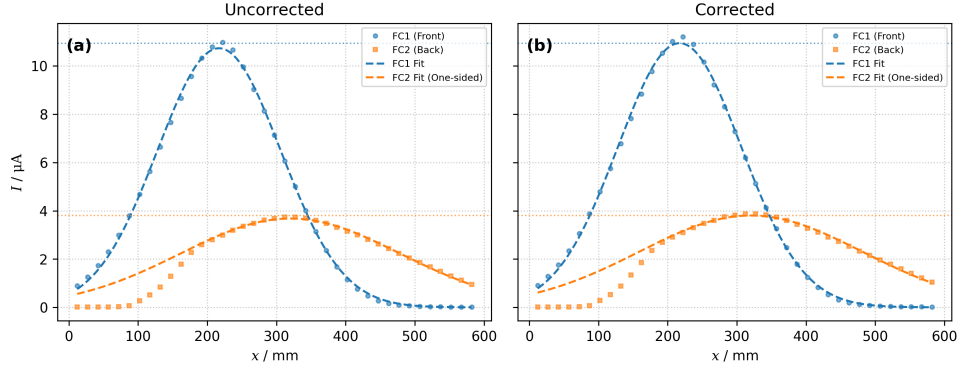


Figure 13: This Figure compares the beam profiles for the operating point at $V_{\text{NHV}} = 750$ and $V_{\text{PHV}} = 200$ with (b) and without (a) divergence and CEX correction applied. The dotted lines help to spot the minor visual difference in peak height. Although the differences appear small, they have a meaningful impact on the final divergence calculation.

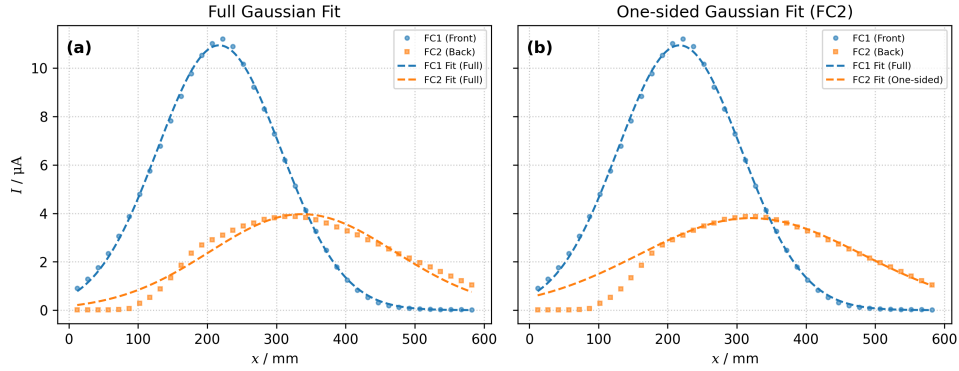


Figure 14: These two plots represent the same experimental data points at $V_{\text{NHV}} = 750$ and $V_{\text{PHV}} = 200$, but analyzed different fitting schemes. For (a), all data points are used for fitting, while for (b), only points ≥ 200 mm are considered. The unphysical shadowing on the left edge of the Gaussian severely changes the shape so that the results are unusable. However, for the one-sided fit, the alignment with all physical data points is great.

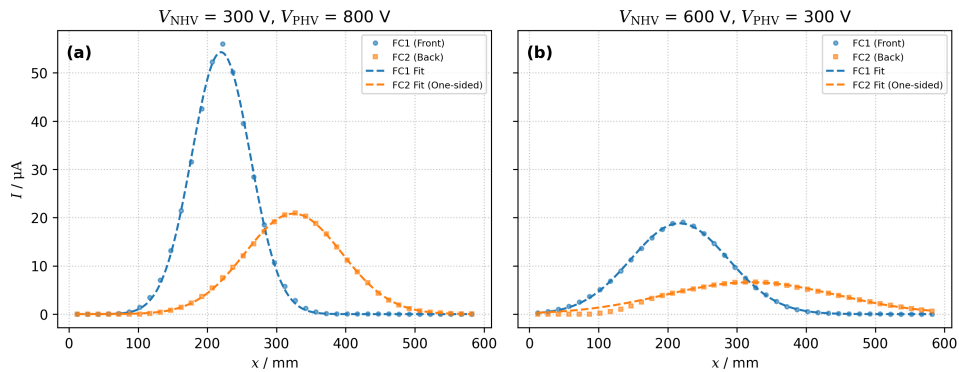


Figure 15: Different operating points (indicated in the subplot titles) cause vastly different beam profiles. (a) depicts a measurement for one of the highest beam energies (800 eV). The Gaussians are sharp, and the divergence is low enough that the shadowing effect on the left side is not visible. In contrast, the 200 eV beam shows a broad Gaussian shape at a much lower peak current density and visible shadowing.

Table 2: The tabulated results for the calibrated RPA measurement. σ_E is the standard deviation for the energy distribution, μ the expectation value calculated by the Gauss fit, E the beam energy with its corresponding error ΔE .

$V_{\text{NHV}} / \text{V}$	$V_{\text{PVH}} / \text{V}$	σ_E / eV	μ / eV	E / eV	$\Delta E / \text{eV}$
-300	1000	22.2	1033.7	999	5
-300	800	13.8	830.1	800	5
-300	600	12.7	623.4	598	4
-450	800	14.7	836.8	806	5
-450	600	11	627.8	602	4
-450	460	11.3	483.7	462	4
-500	400	7	421.2	400	4
-600	800	15.7	837.6	807	5
-600	600	11.5	628.7	603	4
-600	460	11.9	483.4	461	4
-600	300	6.8	317.3	299	4
-750	800	15.6	838.9	809	5
-750	600	11.8	630.6	605	4
-750	300	7.5	316.5	298	4
-750	200	5.1	213	197	4

both corrections are hard to spot visually, their impact on the calculated divergence angle was clearly visible during implementation.

At smaller x positions, the FC2 beam profile exhibits a clear abnormality (see Fig. 14). The ion current decreases prematurely on the left side of the Gaussian, starting at around 200 mm. This effect is caused by the fact that FC2 sits inside the shadow of the RPA and FC1 assemblies that are mounted diagonally in front of FC2. For increasing divergence angles (i.e. lower PHV voltages), the ion trajectories that would enter FC2's aperture impact with the structure of the diagnostics and are thus not measurable. The fact that this shadowing effect occurs asymmetrically supports this hypothesis because there is nothing blocking the view to the left of FC2. To account for the shadowing, the Gauss fit for the FC2 profile is only performed for $x > 200$ mm. As a result, the fit perfectly sits on the data points for the unaffected portion of the Gaussian.

4.3 Discussion

The perveance measurements successfully revealed viable operating points that were used for the successive measurements. An overview of all operating voltages is provided in the first two columns in Tab. 2. With PHV voltages ranging from 1000 V to 200 V, a wide range of ion velocities can be utilized during the planned intake measurements. Certain PHV voltages occur multiple times alongside different NHV voltages so that the impact of the NHV voltage on the beam divergence can be studied.

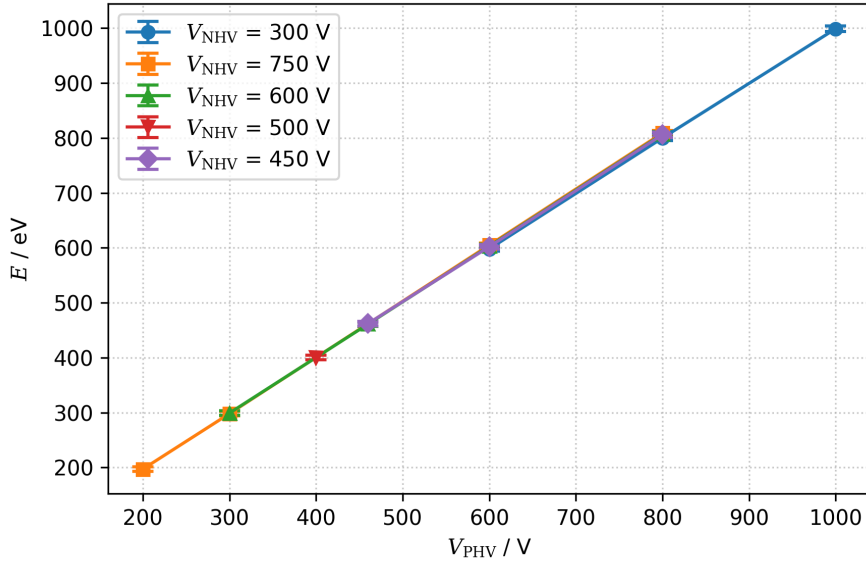


Figure 16: The plot of beam energies against PHV voltage follows a linear trend. This is expected because the NHV voltage should have no effect on the beam energy. The errors bars indicate the error induced by the fit quality and correction uncertainty.

After calibrating the RPA measurements with the empirical calibration curve, the results for the beam energy are depicted in Fig. 16 and also summarized in Tab. 2. As expected, a linear relationship exists between beam energy and PHV voltage. Moreover, the lines for different NHV voltages all overlap, following an almost identical slope. Only for the lowest NHV and highest PHV voltages does the slope diverge slightly from the others. More surprisingly, the relationship closely resembles the angle bisector of the first quadrant, meaning that PHV voltage and beam energy are nearly identical (see Tab. 2). Since the plasma and floating potential bias the plasma positively compared to the screen (PHV) grid, the beam energy would be expected to be slightly elevated. Thus, a systematic error must cause this discrepancy. Possibly, the high voltage power supply for the PHV voltage produces lower voltages than displayed, so that the x-axis is offset by a certain amount. The same could be the true for the power supply used to bias the retarding potential. In future work this should be investigated by comparing the voltage outputs of both power supplies to a calibrated voltmeter. Another possibility is a charge build-up causing a potential difference between the RPA and the thruster. Nevertheless, the RPA results for the beam energy validate the approximate ion energy before impacting the intake. For future measurements the described discrepancy should be reviewed closely. However, for the input parameters of the comparative simulations the relative uncertainty (max. 5 %) will only have a minor influence on the results.

After all post-processing steps for the beam profile measurements are completed, the results of the Gauss fits can be used to evaluate the beam divergence. For this, the 2σ width, that contains 95 % of the beam particles, is compared between the two beam profiles. Over the distance $\Delta y = 218 \text{ mm}$ that the ion beam travels between

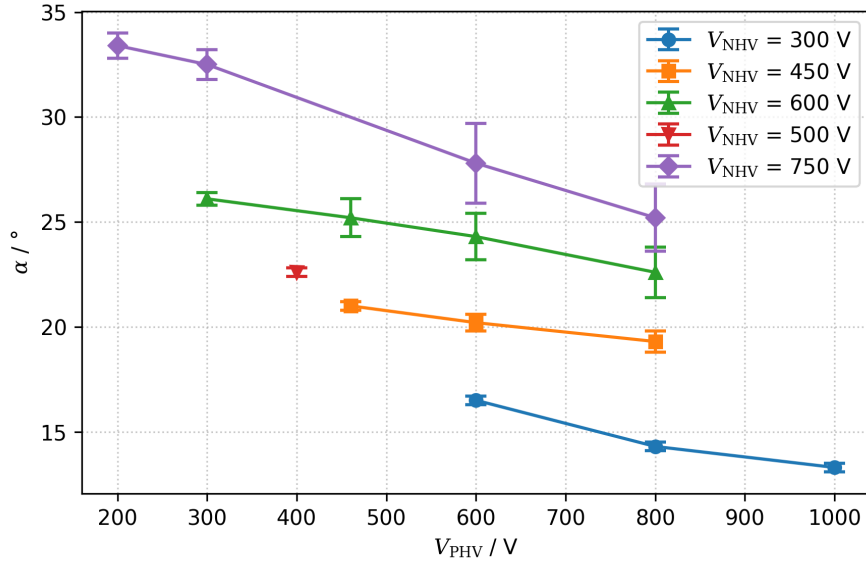


Figure 17: This graph presents the results for the calculated beam divergence. For a constant NHV voltage, the divergence increases almost linearly towards smaller PHV voltages. The slope for this increase is most moderate for the NHV voltages in the center. The ideal operating points lie towards the bottom left of the curves, providing low beam energies with minimized divergence.

the two measurement planes, the peak broadens on both sides by

$$\Delta d = 2 \cdot (\sigma_2 - \sigma_1). \quad (14)$$

Now, the trigonometric relation between this distance and the separation of the cups is used to determine the divergence angle α as

$$\alpha = \arctan \left(\frac{\Delta d}{\Delta y} \right). \quad (15)$$

Lastly, the results and their corresponding errors are depicted in Fig. 17. As expected, the high NHV voltages that are required to prevent excessive grid erosion at low PHV voltages lead to a significant increase in beam divergence compared to the 10 % to 15 % expected during normal thruster operation.

For the PHV voltages 600 V and 800 V and the lowest two NHV voltages, the divergence error far exceed those of the other operating points. These deviating errors can be attributed to the higher fit uncertainty for the respective measurements. For instance, Fig. 18 portrays the beam profiles and their respective fits for the operating points with the highest divergence errors. It shows that the experimental values do not follow a Gaussian shape but rather have a flattened top. This behavior likely corresponds to processes within the plasma chamber or limitations of the extraction grid system. Since the two depicted profiles were the first two acquired on one of the measurement days, it is possible that the thruster was not yet fully thermalized. A repeated measurement of the beam profiles associated with a large divergence error could help to understand whether this behavior is reproducible.

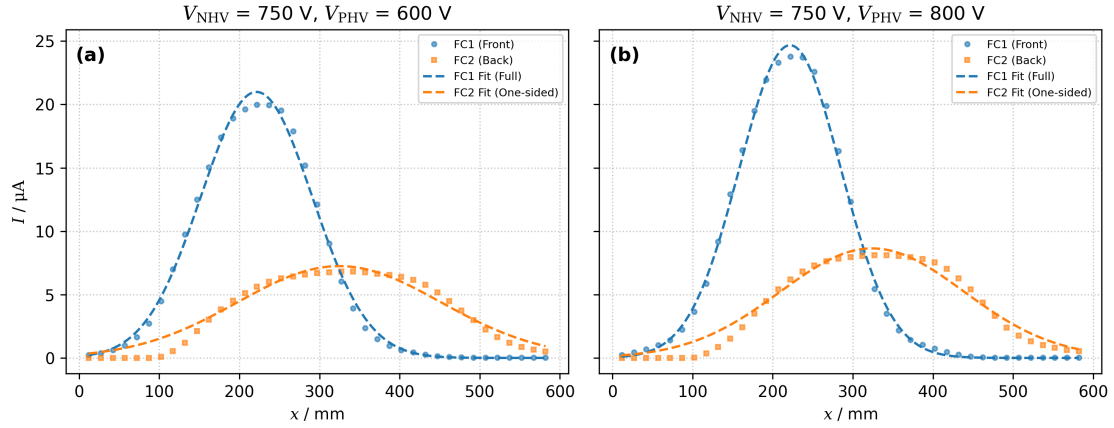


Figure 18: These current profiles correspond to the measurements with the highest divergence errors. For both cups, the profile shapes resemble a Gaussian with a flattened top. This increases the fit uncertainty for the expectation value, further propagating to the divergence error.

5 Summary

This report reviewed the current activities at the ion thrusters working group at the Justus Liebig University dedicated towards research of intake systems for ABEP. After a theoretical introduction, the results of sputtering simulations performed with TRIM were presented. These results highlighted the necessity for minimal ion beam energies to reduce the sputtering yield. Subsequently, two separate experiments were outlined in detail, and the respective beam characterization measurements analyzed and discussed. In the following paragraphs, the most important results are summarized, and future steps outlined.

For the scatter experiment, viable operating points were successfully determined using perveance measurements. For the highest beam energies, two-dimensional beam profile measurements were performed. The linear profiles showed a good alignment of the different components, although the einzel lens axis was slightly misaligned with the thruster and/or aperture system. Still, good focusing with a beam amplification factor larger than 18 was achieved so that the alignment can be used without further adjustments. The angular measurement setup showed an angular acceptance of up to 5° , meaning that only the current-weighted average of the respective distributions has physical significance. This result can be used for future measurements, reducing the number of angular measurement positions and thus decreasing the total measurement time for every operating point. Additionally, the next beam characterization measurements should focus on the low energy operating points where sputtering is minimal, and the ion velocities come closest to orbital speeds.

Concluding with the full intake experiment, the initial step was a perveance measurement to determine viable operating points. For a selection of voltages, beam profile and energy measurements were performed and evaluated. The RPA measurements all produced the characteristic Gaussian curves for the first derivative of the current over voltage data. Using the results from the fits and a linear cal-

ibration curve, the measured voltages were converted to the true beam energies. These energies lie close to or even lower than the respective PHV voltages applied to accelerate the beam. The absence of a positive offset due to the plasma sheath could have several reasons that need to be investigated further in the future. Moreover, the beam profiles were successfully corrected for divergence and CEX effects. Next, the Gauss fit for the second Faraday cup was adjusted such that the shadowed part of the profile is excluded from the fit. Lastly, the divergence of the ion beam was calculated from the standard deviations determined by the Gauss fits. The resulting curves quantify the expected behavior that the lowest beam energies are only reached at the expense of a significant loss of divergence. However, since all operating points were characterized in detail, the corresponding simulations can account for the respective input conditions and still provide a comparison to the experiment. Currently, further beam characterization measurements including charge exchange, Langmuir, and far field measurements are ongoing. In a final step, the well-characterized operating points will be used to perform pressure measurements on the intake and tank structure to determine their performance characteristics. The measurements presented in this report form the foundation for this full intake experiment.

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A Appendix

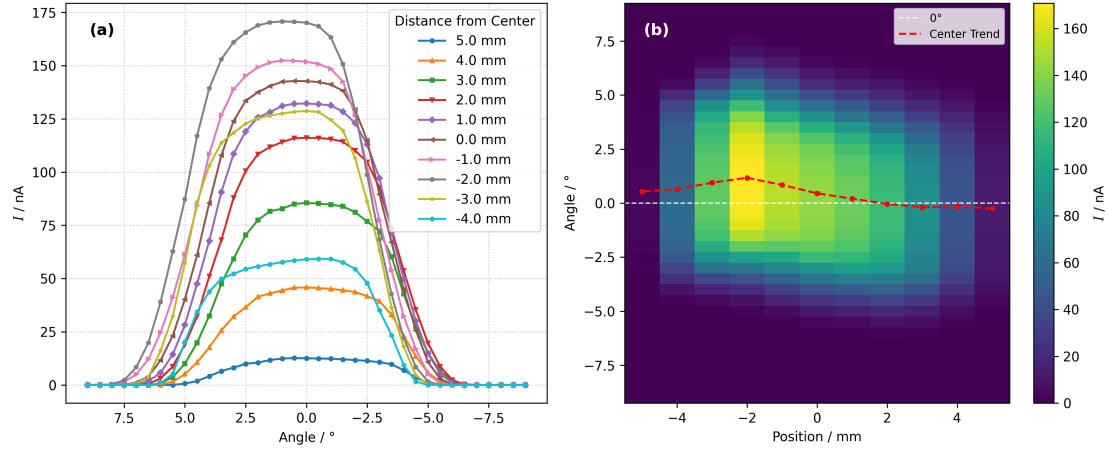


Figure 19: Two-dimensional beam profile measurement for $V_{\text{PHV}} = 800 \text{ V}$ and $V_{\text{NHV}} = 50 \text{ V}$. (a) Angular beam profiles for different positions in the beam. The profiles to the right of the maximum are symmetric, the ones to the left asymmetric. (b) Combined plot depicting trends in angle and position. It shows that the current points increasingly to the left towards the maximum. In accordance with the linear profile the maximum lies about 2 mm left of the center.